

Giordano Alvari

Senior AI / Machine Learning Engineer

Florence, Italy | Remote across Europe | giordano.alvari@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Profile

Senior AI/ML Engineer with 6+ years building production systems for high-stakes enterprise operations. Owns work from business framing and architecture through deployment and evaluation; has led teams of up to five and worked directly with executive stakeholders. Depth across LLM agents and RAG, large-scale predictive ML, hierarchical forecasting, and production platforms on AWS and Kubernetes. Research background in statistical modelling with a peer-reviewed publication in *Genome Biology*.

Experience

Enel — Senior AI Engineer

Jun 2023–Present

Florence, Italy

- Took ownership of an Enel-wide incident-management platform and expanded coverage to approximately **90% of IT incidents**; approximately **5% now close end-to-end without human intervention**.
- Architected and delivered AI-based incident resolution in Italy and Spain: a tool-using agent that executes NOC runbooks against SolarWinds and Huawei/Nokia devices, and a multi-agent RAG platform integrating NetCool, SolarWinds, ServiceNow, memory, and Text2SQL.
- Supported a Seville control room processing approximately **1,000 incidents and 8,000 device diagnostics per month**; established business sign-off for model outputs and worked across Italian, Spanish, and English teams.
- Led a customer-base forecasting programme from concept to production, managing a **team of five** and executive stakeholders. Built hierarchical churn and disconnection forecasts, micro-cluster models, conformal prediction intervals, and price-scenario simulations.
- Shipped a serverless AWS RAG assistant for audit documentation with per-user authorization and source citations; led AI proofs of concept and contributed to group AI governance standards.

Stack: Python, LangChain, LlamaIndex, AWS Bedrock, FastAPI, Pydantic, Kubernetes, pgvector, MongoDB, Spark.

Enel — Machine Learning Engineer

Sep 2020–Jun 2023

Rome, Italy

- Led churn and claim prediction across **18M contracts** on PySpark, delivering a **15% churn reduction and EUR 2.2M in annual savings**; owned ETL, feature engineering, training, and serving.
- Delivered scenario-based churn forecasting, customer segmentation, and electrification cross-selling models; built real-time, edge-deployable anomaly detection for battery energy storage systems.
- Advanced MLOps standards using MLflow, Airflow, and Luigi; taught internal ML courses and won an internal ML competition, earning a job-shadowing week with the global CIO.

Stack: Python, SQL, PySpark, XGBoost, CatBoost, H2O, MLlib, MLflow, Airflow, AWS.

DKFZ / EMBL — Computational Biologist

Nov 2019–Sep 2020

Heidelberg, Germany

- Designed and benchmarked a reproducible single-cell eQTL mapping pipeline across biologically and technically distinct datasets using linear mixed models and neural embeddings; work published in *Genome Biology*.

Selected Projects

- **RadixForge** — C++17 inference server using a radix tree and native llama.cpp APIs to share KV-cache prompt prefixes across concurrent agents on Apple Silicon.
- **Agent Memory Layer** — OpenAI-compatible C++17 proxy for persistent agent memory with local embeddings, SQLite vector search, temporal decay, streaming, and 37 unit/integration tests; benchmarked at up to 11x batch-embedding throughput.
- **global-forecasting** — Python library for global and zero-shot time-series forecasting with lag generation, dynamic/static features, and time-series cross-validation.

Skills

AI/ML: LLM agents, RAG, tool/function calling, Text2SQL, retrieval evaluation, forecasting, conformal prediction, classical ML, PyTorch, scikit-learn, XGBoost.

Engineering: Python, C++17, SQL, FastAPI, Pydantic, PySpark, Docker, Kubernetes, AWS Bedrock, MLflow, Airflow, CI/CD, pgvector, MongoDB, SQLite, llama.cpp.

Leadership: Technical architecture, end-to-end delivery, team leadership, executive communication, AI governance, cross-country stakeholder management.

Education & Publication

MSc, Quantitative and Computational Biology, University of Trento — 110/110 cum laude

2017–2019

BSc, Biomolecular Science and Technology, University of Trento

2014–2017

Publication: Co-author, “Optimising expression quantitative trait locus mapping workflows for single cell studies,” *Genome Biology*, 2021.

Languages

Italian (native) | English (C1) | Spanish (B1) | German (basic)